

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. CHB-503: Organic Chemistry-IV

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer five questions in total, including Question 1 which is compulsory.***Question 1.** Answer all parts of the following: [5 × 2 = 10]

- (a) Why does Michler's ketone not undergo photoreduction to pinacol?
- (b) Differentiate between singlet and triplet carbenes in terms of spin state and geometry.
- (c) Write the product when cyclohexanone is reacted with diazomethane.
- (d) Predict the product: (*E*)-stilbene + singlet carbene (:CH₂) →?
- (e) Why does benzidine rearrangement occur intramolecularly?

Question 2.

- (a) Discuss the stereochemical conformations of 1,2-dimethylcyclohexane and comment on their relative stabilities. [7]
- (b) Discuss Bayer's strain theory and its limitations for higher cycloalkanes. [7]

Question 3.

- (a) Describe the intermolecular and intramolecular insertion reactions of carbenes. [6]
- (b) Predict the correct stability order of the following carbocations, with reasons: benzyl cation, cyclopropylmethyl cation, and isopropyl cation. [4]
- (c) Explain the concept of Kinetic Isotope Effect (KIE) and how it is useful for predicting reaction mechanisms. [4]

Question 4.

- (a) Complete the following reactions, predicting the major product: [3 × 2 = 6]
 - (i) Benzaldoxime $\xrightarrow{\text{Ac}_2\text{O}}$? $\xrightarrow{\text{Conc. H}_2\text{SO}_4}$?
 - (ii) Cyclohexanone $\xrightarrow{\text{HN}_3, \text{H}_2\text{SO}_4}$? (Schmidt reaction)
- (b) Discuss chemical resolution methods for resolving racemic modifications using optically active resolving agents. [4]

- (c) Give the structures of the following: (i) (2*R*, 3*R*)-tartaric acid and (ii) (2*R*, 3*S*)-2, 3-dichlorobutane. [4]